Biology –Strawberry DNA Extraction Lab **Name:_____**
Date:_____Hr:_____

"You will never be able to eat a strawberry again without thinking of how much DNA is in it!"

Background Information:
Ripe strawberries are an excellent source for extracting DNA because they are easy to pulverize and contain enzymes called pectinase and cellulase that help to break down cell walls. And most important, strawberries have eight copies of each chromosome (they are octoploid), so there is a lot of DNA to isolate.

The purpose of each ingredient in the procedure is as follows:

- Dishwasher soap helps to dissolve the cell membrane, which is a lipid bilayer.
- Sodium chloride helps to remove proteins that are bound to the DNA. It also helps to keep the proteins dissolved in the aqueous layer so they don't precipitate in the alcohol along with the DNA.
- Ethanol or isopropyl alcohol causes the DNA to precipitate. When DNA comes out of solution it tends to clump together, which makes it visible.

Materials per student group:
• 1 strawberry
• 10 ml DNA Extraction Buffer (soapy salty water)
• 10 ml ice cold 91% or 100% isopropyl alcohol
• 1 Ziploc TM bag
• 1 clear test tube
• 1 funnel lined with a moistened paper towel

Procedure:
1. Obtain 1 strawberry from the front lab bench.
2. Place strawberry into a Ziploc TM bag and seal shut.
3. Squish for 1 minute to completely squash the fruit.
4. Add 10 ml DNA Extraction Buffer (soapy salty water) and squish for another minute. Try not to make a lot of soap bubbles.
5. Filter the contents of your baggie through a moistened paper towel set in a funnel, and collect the liquid in a test tube. This will take a few minutes. *Do not* squeeze the paper towel. Collect about 3 ml liquid.
6. Add 10 ml of ice cold isopropyl alcohol to the strawberry liquid in the tube. Pour the isopropyl alcohol carefully down the side of the tube so that it forms a separate layer on top of the strawberry liquid.
7. Watch for about a minute. What do you see? You should see a white fluffy cloud at the interface between the two liquids. That's DNA!
8. Rinse your funnel. Put the Ziploc TM bag and paper towel in the garbage.
9. Answer the analysis questions.

BIOLOGY Name_____
DNA Isolation Lab Hour_____Date_____

Analysis Questions

1. A person cannot see a single cotton thread 100 feet away, but if you wound thousands of threads together into a rope, it would be visible at some distance. How is this statement an analogy to our DNA extraction?

2. The strawberry DNA precipitate looks like long and thin fibers, almost like spider webs. Would the DNA look the same if it had come from a human? Explain

3. If you wanted to look at your DNA, could you use the same method of extraction as you did for the strawberries? Why or why not?

4. Why can't you see the double helix?

5. How does your DNA differ from one of your classmate's? How is it similar?

6. How does your DNA differ from the strawberry's DNA? How is it similar?

7. Would the DNA be the same in any cell in the human body? Explain